# Teaching a Neural Network the Glow of the Solar System

A geometrically constrained model of zodiacal light for NASA's SPHEREx mission

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**A geometrically constrained neural network learns the zodiacal light from viewing geometry alone, reaching ~0.02% median error on months it never saw.**

## Background

- **SPHEREx:** NASA space telescope (launched March 2025) mapping the **entire sky in infrared** (0.75–5 μm)
- **Science:** how the universe began, how galaxies formed, where water and other ingredients for life are found

### Zodiacal light

- Sunlight scattered by dust from comets and asteroids, plus the dust's heat glow (Fig. 1)
- Far brighter than the faint glow of distant galaxies SPHEREx aims to measure
- Changes with pointing and Earth's place in orbit → **must be modeled and removed**

### The Kelsall model

- Standard dust model from NASA's COBE; **ZodiPy** evaluates its prediction of the zodiacal-light contribution to what SPHEREx observes

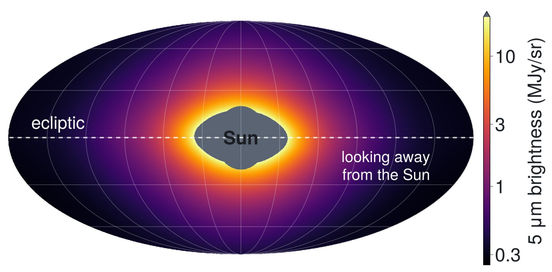



**Figure 1.** The whole sky at 5 μm from SPHEREx (ZodiPy): brightest toward the Sun and along the plane of the planets. Grey: too close to the Sun to observe.

### The problem

- Zodiacal light must be removed from every SPHEREx image, but Kelsall's model uses a finite number of analytic components that don't fully describe the spatial and spectral structure in real data

### Research objective

- A **geometrically constrained** neural network that learns zodiacal light from SPHEREx data and **outperforms the Kelsall model**. First step: ZodiPy's simulated sky

## Methods

### A geometrically constrained model

- No dates or sky coordinates: only **9 numbers describing the viewing geometry** (Fig. 2), dot products of the pointing vector n
- It must learn how brightness depends on geometry, as the physics does, not memorize when and where SPHEREx looked

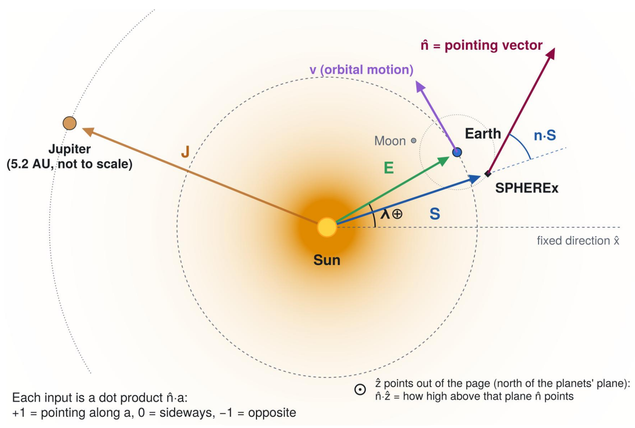



**Figure 2.** The geometry the network sees (top view, not to scale): n = SPHEREx's pointing vector; S, E = SPHEREx, Earth positions; v = orbital motion; λ⊕ = Earth's angle around its orbit. Jupiter and the Moon are tracked for real data: dust clouds leading and trailing Jupiter, and moonlight.

### The 9 inputs

| | |
|---|---|
| **n·S, n·E** | toward or away from the Sun (seen from SPHEREx and from Earth) |
| **n·z** | how far above or below the plane of the planets |
| **n·vS, n·vE** | ahead of or behind the orbital motion (SPHEREx, Earth) |
| **n·x⊕, n·y⊕** | Earth-following frame: x⊕ = Sun → Earth, y⊕ = Earth's motion |
| **sin λ⊕, cos λ⊕** | Earth's position in its orbit: the time of year |

### Training and testing

- 423,607 simulated pointings (140 days, 5 μm); five small networks (Flux.jl), averaged
- Tested on whole months they never saw

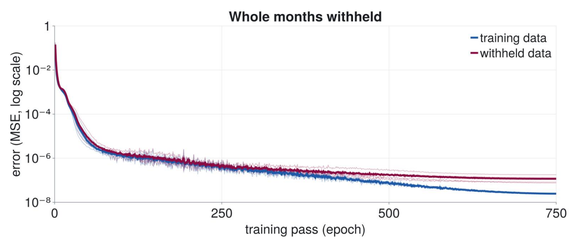



**Figure 3.** Error during training: 5 networks (thin), average (thick). The error on withheld months levels off and never rises: no overfitting.

## Results

**0.018%** median error on months withheld from training

**97.5%** of pointings within 0.1% of ZodiPy

**≤0.03%** median error for every choice of withheld months

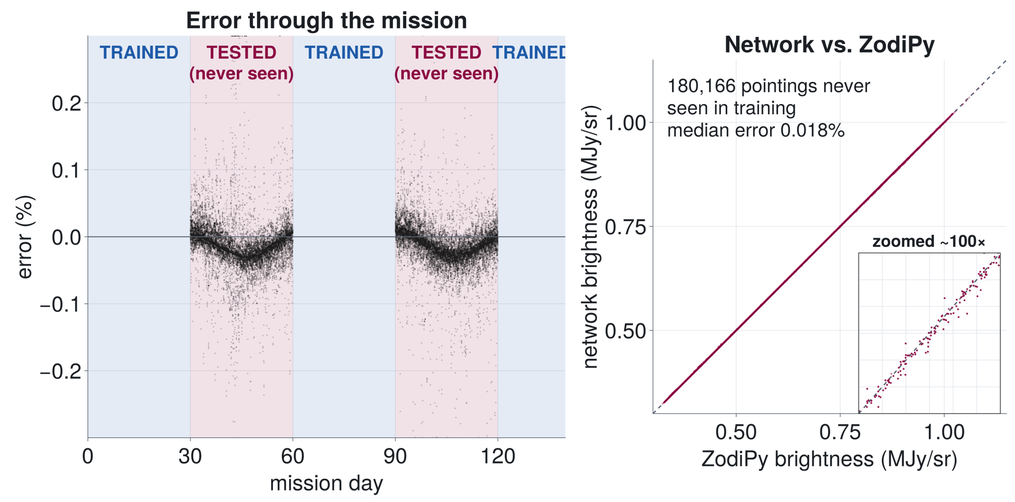



**Figure 4.** Network vs. ZodiPy on the two 30-day blocks never used in training (days 30–60, 90–120; 180,166 pointings). Left: error through the mission; right: every pointing falls on the 1:1 line.

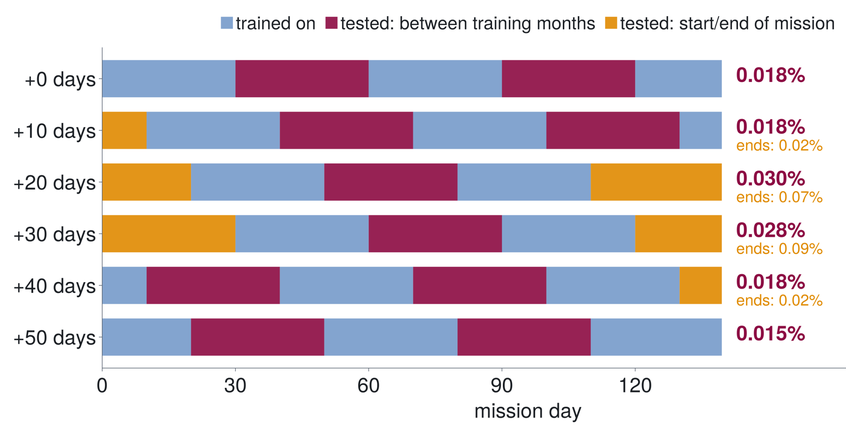



**Figure 5.** Six versions of the test, sliding the withheld months 10 days at a time: all reach 0.015–0.03% (red). Predicting past the start/end of the data (orange) is harder: fine for ~10 days, then errors grow.

- **Largest errors** along the plane of the planets, where the dust bands are; **smallest** toward the poles (Fig. 6)
- **More data:** with all 500 days the whole sky is tested and the plane-of-the-planets error drops (0.083% → 0.064%)

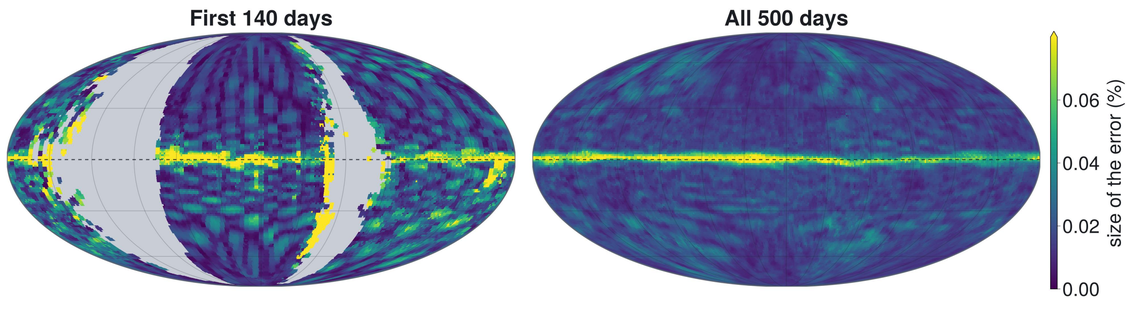



**Figure 6.** Error on pointings the network never saw, across the sky (3.5° patches, about SPHEREx's view). Left: the first 140 days. Right: all 500 days now available; with more than a year of data, Jupiter's direction (n·J) also helps the network tell the two years apart. Grey: not tested.

## Next Steps

**✓ DONE: Validate**
Learn ZodiPy's simulated sky from geometry alone

**IN PROGRESS: Refine**
More data (500 days), longer training, Jupiter's direction

**NEXT: Scale up**
Move training to a computing cluster

**NEXT: Real data**
Train on real SPHEREx images; compare with Kelsall

All results so far are preliminary and use simulated data.

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## References

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